



OOP Project Proposal

Vending Machine Simulation

01286131 OBJECT-ORIENTED PROGRAMMING

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Vending Machine Simulation with Admin and User Modes

Project Description

This project involves developing a C++ program that simulates a vending machine with two operational modes: User Mode (Selling Mode) and Admin Mode. The vending machine will provide functionalities for purchasing items, making payments, and managing stock using SQLite as the database for storing and retrieving necessary data. The program is designed to provide a seamless experience for both end-users and administrators while ensuring robust error handling and logical program flow.

Key Features

- **User Mode (Selling Mode):**
 - **Item Selection:** Users can choose an item from a menu using a unique selection code.
 - **Payment Handling:** Users can insert money in various denominations (100 THB, 20 THB, 10 THB, 5 THB, 1 THB). The program will check if the user has provided enough money.
 - **Stock Management:** After each successful purchase, the stock of the selected item decreases by 1.
 - **Insufficient Payment:** If the payment is insufficient, the system will prompt users to insert more money.
 - **Change Return:** After a successful transaction, the system returns change to the user and updates the stock accordingly.
 - **Out of Stock Handling:** If an item is out of stock, an "OUT OF STOCK" message will be displayed.
 - **Operation Stop Conditions:**
 - The machine stops selling if 50% or more of the items are out of stock.
 - The machine halts if the change box is empty (even with one denomination missing).
 - The vending machine stops if the coin collection box is full.
- **Admin Mode:**
 - **Stock Management:** Admins can set and update the initial stock of any item.
 - **Re-Stocking:** Admins can refill the stock for any item.
 - **Check Financials:** Admins can check the current balance in the change box and the coin collection box.
 - **Money Collection:** Admins can collect money from the coin collection box and refill the change box.

Library

- SQLite3
- C++ Standard Library